

Problem 1

M T W T F S S

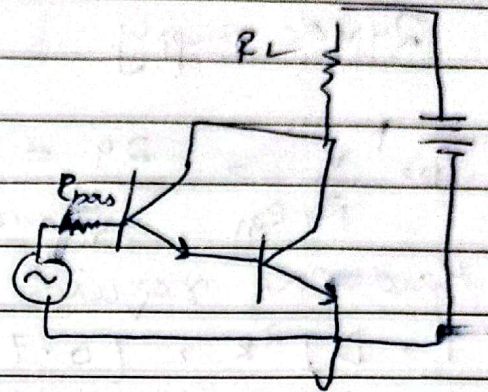
Low side switch 2N3904 to drive 100Ω load powered from 5V dc supply

$$V_{CEsat} = 0.2 \quad V_{BE} = 0.8V$$

$$I_C = \frac{5 - V_{CE}}{R_L} = \frac{5 - 0.2}{100} = 48mA$$

$$I_B = \frac{I_C}{\beta} = \frac{48mA}{20} = 2.4mA$$

$$I_B = 2.4mA$$



High value of e_{in}
2.4V

Low value of e_{in}
0.8V

$$R_{base} = \frac{2.4 - 0.8}{2.4mA}$$

$$R_{base} < 666.67$$

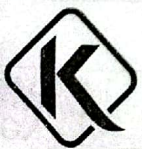
$$D = 0.80$$

$$P_L = D I_C^2 R_L$$
$$= 0.8 (48mA)^2 (100)$$

$$P_L = 184.32m$$

$$P_Q = D V_{CE} I_C$$

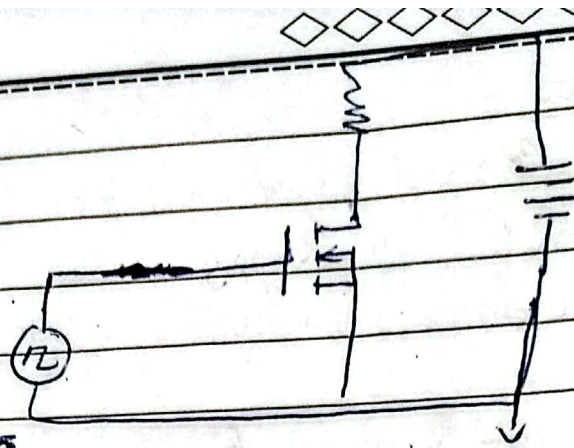
$$P_L = 7.68m$$



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Problem 2

1RF530 35Ω
24Vdc supply



$$\rightarrow I_D = \frac{V}{R_{DS(on)} + R_L} = \frac{24}{35 + 0.14} = 0.6829$$

→ Hard ON require 10V

$$\rightarrow P_L = DI^2R = (0.75)(0.6829)^2(35) = 12.24$$

$$P_T = DI^2R = (0.75)(0.6829)^2(0.14) = 48.9 \text{ mW}$$

$$T_J = T_A + P(\theta_{JC} + \theta_{SA} + \theta_{CS})$$

$$= 60 + 48.9 \text{ m} (62)$$

$$T_J = 63.03 \text{ } \left(\text{No heat sink required} \right)$$

Problem 3

1RF530 35Ω 24V $D = 0.75$ $\theta_{JC} = 62^\circ \text{C/W}$

$$R_{DS(on)} = 0.11$$

$$= \frac{0.11}{4} = 0.0275$$

Voltage drop across parallel transistor

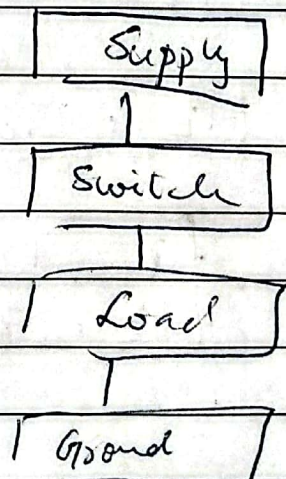
$$V = \frac{0.0275}{0.0275 + 35} \times 24$$

$$V = 19.62 \text{ mV}$$

Power across transistor

$P_T =$

$$P_T = D \frac{V^2}{R} = 0.75 \frac{(19.62 \text{ m})^2}{0.0275} = 10.498 \text{ m}$$



$$P = DV^2$$

$$\overline{R} = 2.61m$$

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$$\begin{aligned} T_J &= T_A + P(\theta_{sc} + \theta_{jc} + \theta_{ga}) \\ &= 60 + 2.61m(62) \\ &= 60.16 \text{ (no heat sink)} \end{aligned}$$

Problem 4 Low side 2N3904 15Vdc $p = 840pF$

$$I_C = \frac{E_{supply} - V_{CE}}{R_L}$$

$$I_{Cmax} = 100mA$$

$$V_{CEon} = 0.3$$

$$V_{BE} = 0.9$$

$$R_L = \frac{15 - 0.3}{100m} = 147\Omega$$

$$\frac{I_B}{I_C} = \frac{100m}{20} = 5mA$$

$$R_{base} \leq \frac{E_{in} - V_{BE}}{I_B} = \frac{9.4 - 0.9}{5m} < 300$$

$$t_{rise} = 2.2 R_L \times C$$

$$= 2.2(150)(840pF) = 277.2nS$$

$$t_{rise} = 277.2nS$$



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Problem 5 $5V$ $4V$ 90% $R_{onT_2} 170^\circ C$

$$R_{on} = 0.3 \times 1.87 = 0.552$$

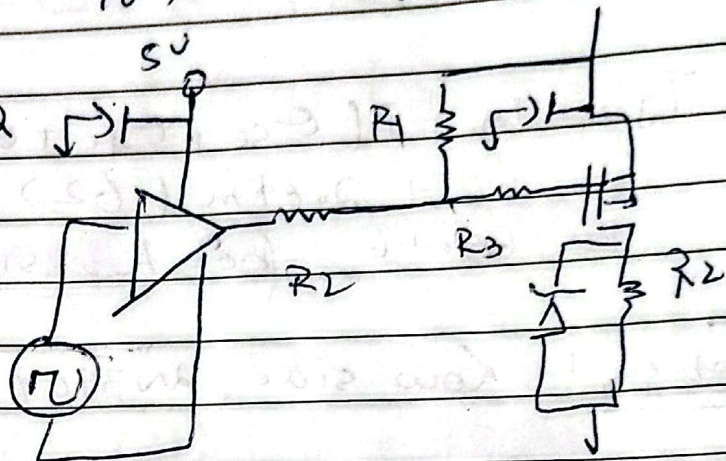
$$R_3 = 10 \Omega$$

$$R_1 + R_2 > 5 - 0.7$$

$$40m$$

$$R_1 + R_2 > 107.5$$

$$R_2 = 0 \quad | \quad R_1 = 150 \Omega$$



$$t_{diss} = 2.2 \times 4 \times 860p = 7.568nA$$

Power dissipation

$$V_L = \frac{4}{4 + 0.552} \times 5 = 4.39$$

$$P_L = D \frac{V_L^2}{R} = 0.9 \frac{(4.39)^2}{4} = 4.336$$

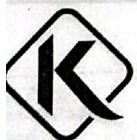
$$P_Q = D \frac{V^2}{R} = 0.9 \frac{(5 - 4.39)^2}{0.552} = 606.68m$$

$$T_J = T_A + P(\theta_{SA} + \theta_{CS} + \theta_{JC})$$

$$\theta_{SA} = T_J$$

$$= 170 + 606.68(1.7)$$

$$= 171.03 \text{ no heat sink}$$

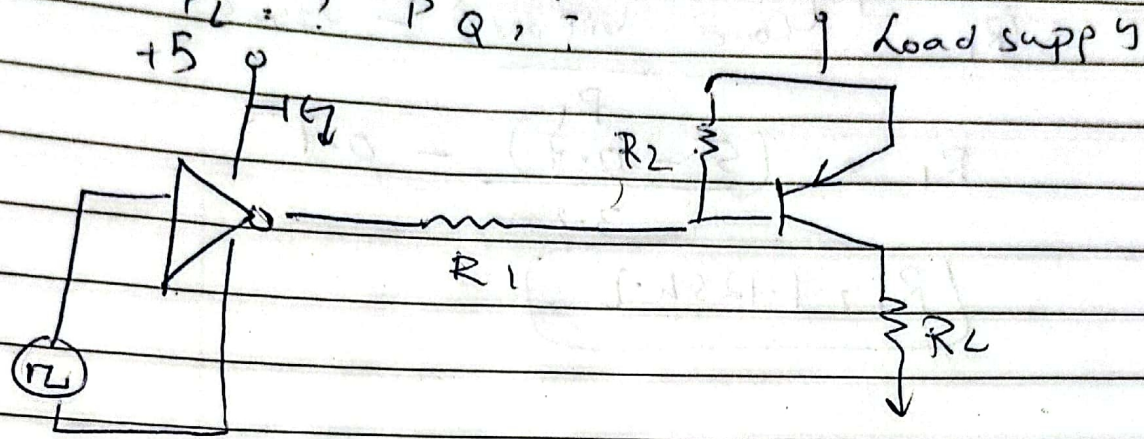


Problem 6

npn high side switch

5V, 50mA R_{load} $D = 0.85$

$P_L = ?$ $P_Q = ?$



$$V_L = 5 - 0.1 = 4.9$$

P_L DVI

$$= 0.85(4.9)(0.05)$$

$$P_L = 0.196$$

$$P_L = 0.196$$

$$P_L = 196mW$$

$$I_{load} = \frac{E_{load} - V_{CE}}{R_{load}}$$

$$V_{load} = E_{load} - V_{CE}$$

$$P_Q = D V_{CE} I_C$$

$$= 0.8(0.1)(0.05)$$

$$P_Q = 4mW$$

Assume $R_2 = 1k\Omega$

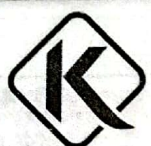
$$I_{R_2} = \frac{V_{BE}}{R_2} = \frac{0.7}{1k} = 0.7mA$$

$$I_{base} > I_{load}$$

$$I_{base} > \frac{I_{load}}{\beta} = 2.5mA$$

R_1

$$I_{R_1} = I_{base} + I_{R_2}$$



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$$= 2.5m + 0.7m$$

$$I_{R_1} = 3.2m$$

$$R_1 = \frac{(E_{load} - V_{BE(on)})}{I_{R_1}} - V_{Z_{40.6}}$$

$$R_1 = \frac{(5 - 0.7)}{3.2m} - 0.7$$

$$R_1 = 1.125k\Omega$$